

6th Grade Team Exam

2025 James Clemens Math Tournament

INSTRUCTIONS:

1. Your team has 60 minutes to complete this exam. Four students from each school will participate in this round.
2. This exam consists of 15 free-response questions, each worth 10 points if answered correctly and 0 points if answered incorrectly or left unanswered.
3. Calculators, books, and other aids are prohibited during this examination. Scratch paper will be provided for calculations. Diagrams are not necessarily drawn to scale.
4. Mark your answers to the questions in the provided team answer sheet. You may use the test booklets for scratch work, but only answers marked in the team answer sheet will be counted. If you require additional scratch paper, simply raise your hand and the room proctor will assist you.
5. All answers for the team test will be a positive integer from 1 to 999.
6. A team's final score will be calculated by adding each school's top 4 written test scores per grade (excluding tie breaking questions) as well as the team round score, for a maximum of 550 points.
7. In the event of a tie, the team with the highest team round score will be favored. If team round scores result in another tie, answers will be evaluated starting backwards from question 15 to 1 to determine a winner.
8. Although this math tournament is intended to demonstrate your knowledge and skills in math, it is also a great opportunity for you to interact with your fellow peers, so be sure to enjoy yourself and have fun!
9. This test will be taken in teams of 4, with unofficial teams being able to compete (i.e., groups can take it for fun).

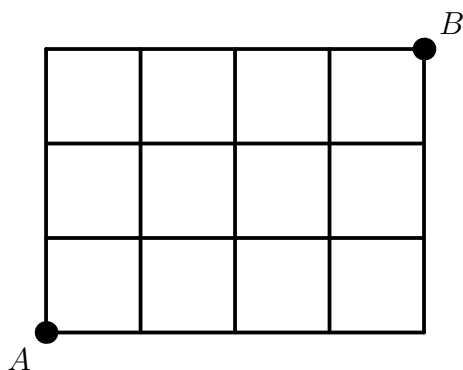
1. The ratio of the pennies I have to the pennies my friend has is $3 : 2$. I decide to give my friend 12 pennies, and now we have the same amount of pennies. How many pennies did I have originally?
2. Harshtha buys a \$200 bicycle at Walmart. If Walmart charges a 5% sales tax, what is the total amount of money she has to pay, in dollars? (You may leave out the dollar sign in your answer.)
3. A recipe for lemonade calls for 2 cups of lemon juice, 7 cups of water, and 1 cup of sugar to make 10 cups of lemonade. How many cups of water do 5 gallons of lemonade require? (1 gallon = 16 cups)
4. Farmer Fred has chickens and horses on his farm. He counts a total of 80 heads and 194 feet. How many chickens are on his farm?
5. Bob has 60 apples. He wants to put these apples into some number of piles, with the same number of apples in each pile and no apples left over. How many different ways can he do this?
6. Sam is traveling between cities A and B. He starts at city A and drives his car to city B, in a straight line. Once he gets to city B, he immediately turns around and drives back to city A along the same route. If Sam's average speed going to city B is 75 mph, and his average speed coming back to city A is 50 mph, then what is his average speed for the entire trip, in miles per hour?

7. The numerical value of the perimeter and the area of an equilateral triangle are the same. If the side length of the triangle is x , then what is x^2 ?
8. When a positive integer x is divided by 4, the remainder is 3. When x is divided by 5, the remainder is 4. When x is divided by 6, the remainder is 5. What is the least possible value of x ?
9. Jeffrey has just started learning fractions, and he writes down the (incorrect) equation

$$\frac{3}{8} = \frac{21}{31}.$$

Erik tells him that although the equation is incorrect, if he adds the same positive integer to the numerator and denominator of both fractions, then the equation will be correct. What is the number to be added?

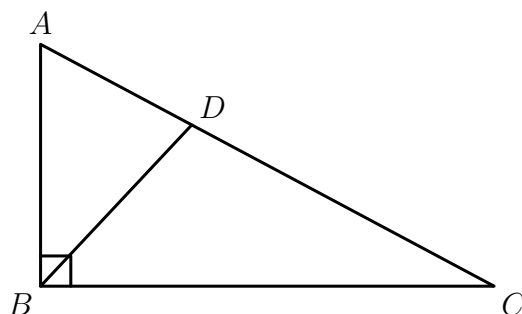
10. A mouse starts from point A on the 3×4 grid below and crawls to point B along the lines. It can only crawl rightward and upward.



How many possible paths are there?

11. Vineet ate a total of 169 cookies in 13 days. Each day, he ate 2 more cookies than the previous day. How many cookies did he eat on the last day?

12. In the figure below, point D is on $\overline{AC}$ such that $\frac{AD}{CD} = \frac{1}{2}$.



Given that the area of triangle ABC is 180 cm^2 , find the area of triangle BCD .

13. A fair coin is flipped 9 times. The probability that less than 5 of the flips will land heads up can be expressed as a simplified common fraction in the form $\frac{a}{b}$. Answer with $b - a$.

14. The eight blanks in the following arithmetic expression are to be filled with the digits 1 through 8, inclusive, so that the maximum possible product of the two 4-digit numbers is achieved. Each digit may be used only once.

____ _ × ____ _

What is the sum of the digits of the larger number?

15. A 1×3 enemy spaceship is going to land on a 5×5 field, taking up 3 of the 25 unit squares on the field. Mines are placed in some of the unit

squares, and if the spaceship lands on a mine, it will explode. Determine the minimum number of mines needed such that there exists a way to place the mines so that the spaceship is guaranteed to explode, no matter where it lands.